ENSFLOW: Self-Regulating Heteroscedastic Bridges on $\mathrm{SE}(3)$ for RNA Conformational Ensemble Generation

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Abstract

Modeling the conformational ensembles of RNA is a central open problem in machine learning for structural biology: RNA function often arises from multi-state flexibility rather than a single native fold, yet every modern structure-prediction system, including the RNA branch of AlphaFold3, outputs a single point estimate. We introduce ENSFLOW, a three-level hierarchical framework for RNA ensemble generation on $\mathrm{SE}(3)^L$ built around a *conditional heteroscedastic stochastic bridge*. The central object is a residue-wise, state-dependent bridge width field whose modulation is learned jointly with an $\mathrm{SE}(3)$ flow-matching drift. Our key methodological contribution is a *self-regulating* training objective: we supervise the bridge width via a heteroscedastic Gaussian negative log-likelihood of the drift model's own residual, and prove that its closed-form optimum is $\sigma^* = \|r\|$—the bridge width at any (t, i) is exactly the drift model's residual magnitude at that point. This yields a temperature-free, self-calibrating uncertainty signal that ties stochastic exploration directly to model reliability, and admits a Bayesian interpretation as the trace of the posterior covariance of the clean state given the noisy state. We further introduce *B-factor anchoring*, a supervision signal that aligns the learned uncertainty field with experimentally measured atomic displacement parameters—to our knowledge the first direct link between ML epistemic uncertainty and crystallographic aleatoric flexibility in an $\mathrm{SE}(3)$ generative model. We prove that Level 1 (deterministic flow matching) and the fixed-width stochastic bridge both arise as strict degenerate limits of Level 2, giving a clean hierarchy. **[TODO: Empirical results on RNA ensemble benchmarks.]**

1 Introduction

RNAs are not rigid. Unlike most proteins, where a single "native" fold captures the dominant functional state, many RNAs function precisely through conformational heterogeneity: riboswitches undergo ligand-induced structural switches [Serganov and Nudler, 2013], ribozymes traverse distinct catalytic states, and regulatory RNAs adopt transient interaction-competent folds [Mustoe et al., 2014]. Representing an RNA with a single structure fundamentally misses this functional machinery. Yet every modern structure prediction system—including the RNA module of AlphaFold3 [Abramson et al., 2024] —outputs a single point estimate, leaving conformational *ensemble* generation as one of the most consequential open problems in machine learning for RNA structural biology.

What existing $\mathrm{SE}(3)$ generative models miss. Recent progress on $\mathrm{SE}(3)$-equivariant generative models for biomolecular backbones has been rapid. FrameDiff [Yim et al., 2023b] and FrameFlow [Yim et al., 2023a] established $\mathrm{SE}(3)$ diffusion and flow matching for protein backbones; FoldFlow and its stochastic variant FoldFlow-SFM [Bose et al., 2024] extended flow matching to stochastic bridges on $\mathrm{SE}(3)$ and observed that stochasticity improves novelty and diversity. For RNA, RNA-FrameFlow [Anand et al., 2024] adapted this to nucleotide frames. Ensemble-focused methods—Distributional Graphormer (DiG) [Zheng et al., 2024], BioEmu [Lewis et al., 2024], AlphaFlow [Jing et al., 2024], and Str2Str [Lu et al., 2024]—apply similar ideas to *protein* ensembles. Strikingly absent from all of these: any mechanism to learn *where* and *how much*

stochasticity to inject. FoldFlow-SFM uses a global time-dependent schedule $\sigma(t)$; DiG and BioEmu use fixed diffusion coefficients; AlphaFlow reuses a flow-matching schedule with no local adaptation. The stochasticity is global and state-agnostic—the same amount of noise at every residue regardless of whether that residue sits in a stable stem or a flexible loop.

The key insight. RNA flexibility is emphatically not global. A hairpin has rigid base-paired stems and flexible apical loops; a riboswitch aptamer is stable while its expression platform is dynamic; many tRNAs have one stable core and one mobile anticodon arm. Uniform noise erases this signal. What we want is a *learned heteroscedastic uncertainty field* on $\mathrm{SE}(3)^L$ that assigns local stochasticity where the model itself is uncertain, not a single global temperature. This raises two questions:

(i) What training signal should guide such a field?

(ii) Does the learned field correspond to anything physically measurable?

We answer both. *First*, we show that the bridge width maximizing the likelihood of the drift model's own residual is the residual magnitude itself—yielding a closed-form, temperature-free, self-regulating objective (§4.3, Prop. 1). This ties stochastic exploration to model reliability: *the model is more stochastic exactly where it is less certain about the clean structure*. *Second*, we show that supervising the learned bridge width with experimental B-factors (crystallographic atomic displacement parameters) aligns it with physically measured flexibility, giving a concrete, falsifiable connection between model epistemic uncertainty and experimental aleatoric flexibility (§4.5).

Contributions.

1. **EnsFlow framework.** A three-level hierarchical model for RNA ensemble generation on $\mathrm{SE}(3)^L$: Level 1 (deterministic flow matching), Level 2 (conditional heteroscedastic stochastic bridge), Level 3 (bridge with B-factor anchoring). We prove that Level 1 and fixed-width FoldFlow-SFM-style bridges arise as strict degenerate limits of Level 2 (Prop. 3).
2. **Self-regulating NLL objective.** A heteroscedastic Gaussian NLL loss on the drift residual whose closed-form pointwise minimum is $\sigma^* = \|r\|$ (Prop. 1). Under a Bayes-optimal drift assumption, this corresponds to the trace of the posterior covariance of the clean state given the noisy state (Cor. 1). No temperature, no KL weight, no hand-tuned balance between exploration and fidelity.
3. **B-factor physical anchoring.** A novel supervision signal aligning the learned bridge width with experimentally measured B-factors, providing the first direct link between generative-model epistemic uncertainty and physical flexibility measurements in an $\mathrm{SE}(3)$ framework.
4. **Empirical validation.** On **[TODO: RNA ensemble benchmarks]**, we show **[TODO: improvements in]** ensemble coverage, diversity, and physical validity over deterministic and fixed-width bridge baselines, with the learned uncertainty field exhibiting significant Spearman correlation against held-out B-factors—even when B-factor supervision is disabled (Level 2).

2 Related Work

Table 1 places EnsFlow against representative prior work along the axes that matter for our contribution.

$\mathrm{SE}(3)$ **flow/diffusion for biomolecules.** FrameDiff [Yim et al., 2023b], FrameFlow [Yim et al., 2023a], Chroma [Ingraham et al., 2023], Genie [Lin and AlQuraishi, 2023], and RFdiffusion [Watson et al., 2023] place equivariant generative models directly on $\mathrm{SE}(3)^L$ using either diffusion or flow matching. All use global, fixed noise schedules. RNA-FrameFlow [Anand et al., 2024] is the closest single-structure antecedent for our target domain; EnsFlow builds on its backbone architecture and $\mathrm{SE}(3)$ flow-matching drift.

Table 1: Positioning of ENSFLOW against representative prior work on $\mathrm{SE}(3)$ generative modeling and ensemble generation. "SDE" = stochastic dynamics; "Learned σ" = noise coefficient learned jointly with drift; "State-dep." = noise depends on current noisy state; "Residue-wise" = noise varies per residue; "Physical anchor" = learned uncertainty explicitly tied to experimentally measured flexibility. ENSFLOW is the first to check all boxes.

Method	SDE?	Learned σ	State-dep.	Residue-wise	Phys. anchor	Target
FrameDiff [Yim et al., 2023b]	diffusion	fixed	–	–	–	protein
FrameFlow [Yim et al., 2023a]	ODE	–	–	–	–	protein
RNA-FrameFlow [Anand et al., 2024]	ODE	–	–	–	–	RNA
FoldFlow-SFM [Bose et al., 2024]	✓	fixed $\sigma(t)$	–	–	–	protein
AlphaFlow [Jing et al., 2024]	≈	fixed schedule	–	–	–	protein ens.
DiG [Zheng et al., 2024]	diffusion	fixed	–	–	–	protein ens.
BioEmu [Lewis et al., 2024]	ODE/SDE	fixed	–	–	–	protein ens.
Str2Str [Lu et al., 2024]	diffusion	fixed	–	–	–	protein ens.
Improved DDPM [Nichol and Dhariwal, 2021]	diffusion	✓(VLB)	–	iso. pixel	–	images
EnsFlow (ours)	✓	✓(NLL)	✓	✓	✓	**RNA ens.**

Stochastic bridges and interpolants. FoldFlow-SFM [Bose et al., 2024] replaces deterministic flow paths with $\mathrm{SE}(3)$ stochastic bridges, using translation Brownian bridges and SO(3) IGSO(3) approximations. Stochastic interpolants [Albergo et al., 2023] and Schrödinger bridge matching [Shi et al., 2023] provide the broader theoretical frame. In all of these, the diffusion coefficient is a fixed function of t—never residue-wise and never state-dependent. Our Level 2 is a strict generalization (Prop. 3).

Ensemble generation for biomolecules. DiG [Zheng et al., 2024], BioEmu [Lewis et al., 2024], AlphaFlow [Jing et al., 2024], and Str2Str [Lu et al., 2024] generate protein conformational ensembles, typically trained on MD trajectories or multi-state experimental data. RNA-specific ensemble generation has received much less attention; we are not aware of a published ML method targeting RNA conformational ensembles with residue-resolved flexibility.

Learned variance in generative modeling. Improved DDPM [Nichol and Dhariwal, 2021] learns variance via a hybrid $L_{\text{simple}} + L_{\text{vlb}}$ loss; Variational Diffusion Models [Kingma et al., 2021] learn an SNR schedule. Both operate on Euclidean pixels and learn a single time-indexed variance shared across all coordinates. Our per-residue, state-dependent σ on $\mathrm{SE}(3)^L$, trained by a simple NLL with closed-form minimum at $\sigma^* = \|r\|$, is a different animal.

3 Preliminaries

Nucleotide frames. Following Anand et al. [2024], we represent an RNA backbone of length L as a sequence of rigid frames $G = \{G_i\}_{i=1}^L$, $G_i = (x_i, R_i) \in \mathrm{SE}(3)$, where $x_i \in \mathbb{R}^3$ is a reference atom position (C4′ for RNA) and $R_i \in \mathrm{SO}(3)$ is a local frame constructed from neighboring atoms. Given sequence $a = (a_1, \ldots, a_L)$, the generation task is to model $p(G \mid a)$.

Time convention. Throughout, $t = 0$ denotes the noise/prior end and $t = 1$ denotes the clean data end, so at training time we observe a clean (x_1, R_1) and sample a noisy intermediate (x_t, R_t). The drift of a flow matching model predicts the *clean endpoint* $\hat{G}_{1|t} = (\hat{x}_{1|t}, \hat{R}_{1|t})$; the velocity is induced as

$$u^x(i,t) = \frac{\hat{x}_{i,1|t} - x_{i,t}}{1-t}, \qquad u^\omega(i,t) = \frac{1}{1-t}\,\mathrm{Log}\big(R_{i,t}^\top \hat{R}_{i,1|t}\big), \tag{1}$$

where $\mathrm{Log} : \mathrm{SO}(3) \to \mathfrak{so}(3) \cong \mathbb{R}^3$ is the matrix logarithm (rotation-vector) map.

$\mathrm{SE}(3)$ **flow matching.** The Level 1 drift is trained by the standard flow-matching MSE against the ground-truth velocity field:

$$\mathcal{L}_{\text{flow}} = \mathbb{E}_{t,G_1,G_t}\left[w_x\left\|u_\theta^x - u^{x,*}\right\|_2^2 + w_\omega\left\|u_\theta^\omega - u^{\omega,*}\right\|_2^2\right], \tag{2}$$

with weights $w_x, w_\omega > 0$ balancing translation (Å) and rotation (rad).

$\mathrm{SO}(3)$ **Brownian bridges via tangent-space Gaussians.** For small angular variance, the IGSO(3) density of a rotation R with concentration σ^2 is well-approximated by the tangent-space Gaussian $p(R \mid \sigma^2) \propto \exp\big(-\|\mathrm{Log}(R)\|^2/(2\sigma^2)\big)$; the relative error in log-likelihood vanishes as $\sigma \to 0$ [Nikolayev and Savyolov, 1997, Leach et al., 2022]. We use this approximation in §4.3.

4 Method

Our central object is a conditional heteroscedastic stochastic bridge on $\mathrm{SE}(3)^L$. We first formalize the bridge and its parameterization (§4.2), then introduce the self-regulating NLL objective (§4.3), the BridgeWidthNet that implements the modulation field (§4.4), and the B-factor anchor (§4.5). We conclude with the theoretical properties of the framework (§4.6). Figure 1 gives a bird's-eye view of the training-time data flow: a single flow-matching drift network shares its outputs both with the flow loss and with a lightweight BRIDGEWIDTHNET that predicts the per-residue stochastic bridge width.

4.1 Problem formulation

Let $p^*(G \mid a)$ be the target distribution over RNA backbone conformations given sequence a. For a flexible RNA, $p^*(\cdot \mid a)$ is multi-modal: distinct structural states correspond to distinct functional configurations. A single-structure predictor collapses this distribution to a point estimate $\mathbb{E}[G \mid a]$ or mode estimate $\arg\max p^*(\cdot \mid a)$, losing the multi-state information. Ensemble generation instead seeks a generator $q_\theta(G \mid a)$ whose samples cover the support of $p^*(\cdot \mid a)$. Faithful coverage requires *local* stochasticity aligned with the true per-residue flexibility of p^*.

4.2 Level 2: Conditional heteroscedastic $\mathrm{SE}(3)$ stochastic bridge

The SDE. Given a drift u_θ parameterized as in Eq. (1) and a residue-wise bridge width field $\sigma_i(t, G_t, a)$, we define the generative process on $\mathrm{SE}(3)^L$ by the coupled SDE

$$dx_{i,t} = u_\theta^x(i,t)\,dt + \sigma_i^x(t, G_t, a)\,dW_{i,t}^x, \tag{3}$$

$$dR_{i,t} = R_{i,t}\mathrm{Exp}\Big(u_\theta^\omega(i,t)\,dt + \sigma_i^\omega(t, G_t, a)\,dW_{i,t}^\omega\Big), \tag{4}$$

where $W_{i,t}^x$ is standard Brownian motion in $\mathbb{R}^3$ and $W_{i,t}^\omega$ is isotropic noise in the Lie algebra $\mathfrak{so}(3)$, and $\mathrm{Exp} : \mathfrak{so}(3) \to \mathrm{SO}(3)$ is the matrix exponential. The crucial point is that $\sigma_i^x, \sigma_i^\omega \in \mathbb{R}_{\geq 0}$ are *per residue*, depend on the entire current state G_t and on the sequence a, and are *learned jointly* with the drift.

Endpoint-safe base schedule and bounded modulation. We decompose each width into a fixed endpoint-vanishing base schedule and a learned, bounded modulation factor:

$$\sigma_i^x(t, G_t, a) = \sigma_0^x(t) \cdot \alpha_i^x(G_t, a, t), \quad \sigma_i^\omega(t, G_t, a) = \sigma_0^\omega(t) \cdot \alpha_i^\omega(G_t, a, t), \tag{5}$$

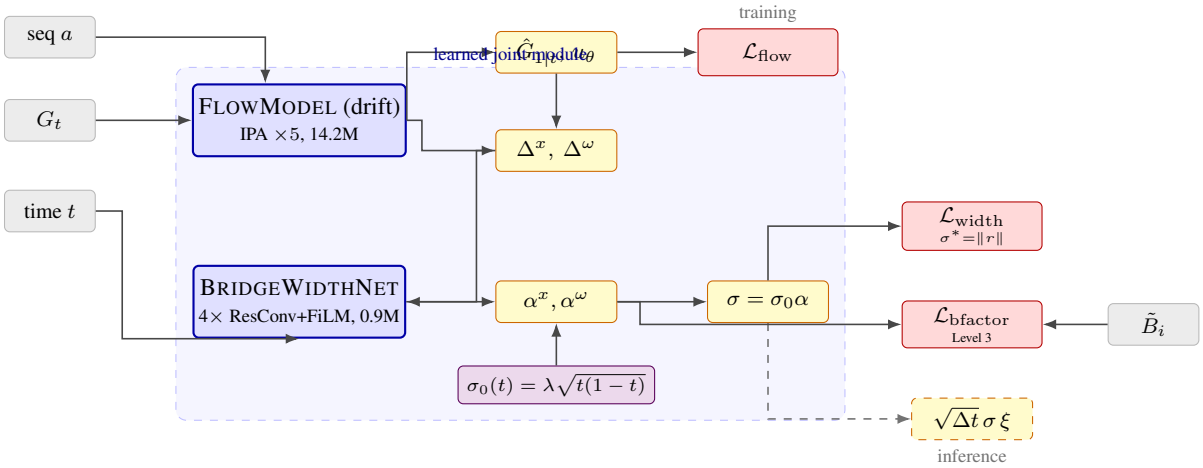

Figure 1: **EnsFlow architecture and training data flow.** Two networks share the sequence/state input. The **FlowModel** drift (top blue, 14.2M) is the standard $\mathrm{SE}(3)$ flow-matching backbone: it predicts the clean endpoint $\hat{G}_{1|t}$ and the induced velocity u_θ, supervised by the flow-matching loss $\mathcal{L}_{\text{flow}}$. From the drift's own prediction and the noisy state we compute per-residue geometric residuals $\Delta^x = x_t - \hat{x}_{1|t}$ and $\Delta^\omega = \mathrm{Log}(\hat{R}_{1|t}^\top R_t)$. The **BridgeWidthNet** (bottom blue, 0.9M) consumes these residuals together with the drift's single/pair embeddings (carried inside the drift arrow) and the timestep t, and outputs bounded per-residue modulation factors $\alpha^x, \alpha^\omega \in [\alpha_{\min}, \alpha_{\max}]$. Multiplied by the endpoint-vanishing base schedule $\sigma_0(t) = \lambda\sqrt{t(1-t)}$ they yield the effective bridge width σ. Level 2 trains σ via the heteroscedastic NLL loss $\mathcal{L}_{\text{width}}$, whose pointwise minimum is $\sigma^* = \|r\|$ (Prop. 1). Level 3 further anchors α to normalized experimental B-factors $\tilde{B}_i$ via $\mathcal{L}_{\text{bfactor}}$. At inference (dashed path), the same σ drives the Euler–Maruyama noise term $\sqrt{\Delta t}\,\sigma\,\xi$. All residuals and embeddings feeding BridgeWidthNet are *detached* from the drift computation graph, so bridge-width gradients do not flow back into the drift network.

with

$$\sigma_0^x(t) = \lambda_x\sqrt{t(1-t)}, \qquad \sigma_0^\omega(t) = \lambda_\omega\sqrt{t(1-t)}, \tag{6}$$

and modulations

$$\alpha_i^{\cdot} \;=\; \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot \sigma_{\text{sig}}(u_i), \qquad u_i = \text{(network output)}, \tag{7}$$

where σ_{sig} denotes the logistic function. By construction, $\sigma_0^{\cdot}(0) = \sigma_0^{\cdot}(1) = 0$, so the diffusion coefficient vanishes at both endpoints regardless of α; the bounded modulation prevents σ from either collapsing to zero or diverging.

Why this parameterization. The decomposition (5)–(7) has three properties we will use repeatedly in §4.6: (a) endpoint preservation is free (Prop. 2); (b) the Level 1 ODE limit is recovered strictly by $\lambda \to 0$ (Prop. 3(a)); (c) the fixed-width FoldFlow-SFM-style bridge is recovered strictly by $\alpha \equiv 1$ (Prop. 3(b)). Novelty lives entirely in how we *learn* α, which we now describe.

4.3 Self-regulating training via heteroscedastic NLL

The drift u_θ is trained with the usual flow-matching loss (2); the bridge width field is trained by a *heteroscedastic Gaussian NLL* on the drift model's own residuals. We first state the construction, then the proposition that justifies it.

Residual observations. At each training step, we sample $t \sim \mathrm{Uniform}(0,1)$ and a clean target $G_1 = (x_1, R_1)$, form a noisy intermediate G_t by the standard flow matching interpolation, and compute the drift's clean-endpoint prediction $\hat{G}_{1|t} = (\hat{x}_{1|t}, \hat{R}_{1|t})$. Define the per-residue residual *norms*

$$r_i^x(t) \;=\; \big\|x_{i,1} - \hat{x}_{i,1|t}\big\|_2, \qquad r_i^\omega(t) \;=\; \big\|\mathrm{Log}\big(\hat{R}_{i,1|t}^\top R_{i,1}\big)\big\|_2. \tag{8}$$

These are detached from the drift computation graph (so that bridge-width gradients do not affect the drift) and passed as observations to the heteroscedastic NLL.

Heteroscedastic 1D NLL. Under the working assumption that $r_i^{\cdot}$ is drawn from a zero-mean 1D Gaussian with unknown variance σ^2, its negative log-likelihood is

$$\ell(r,\sigma) \;=\; \frac{r^2}{\sigma^2} + \log\sigma^2 + \text{const}, \tag{9}$$

which we minimize jointly over the residue indices and modalities:

$$\mathcal{L}_{\text{width}} \;=\; \frac{1}{w_x + w_\omega}\,\mathbb{E}\Bigg[w_x \sum_i \Big(\tfrac{(r_i^x)^2}{\sigma_{x,i}^2} + \log\sigma_{x,i}^2\Big) + w_\omega \sum_i \Big(\tfrac{(r_i^\omega)^2}{\sigma_{\omega,i}^2} + \log\sigma_{\omega,i}^2\Big)\Bigg]. \tag{10}$$

Here $\sigma_{x,i} = \sigma_0^x(t)\cdot\alpha_i^x$ etc., and the per-modality weights w_x, w_ω are reused from the Level 1 flow-matching loss (Eq. 2) to balance the mismatched units of translation (Å) and rotation (rad) without introducing a new hyperparameter.

Rotation NLL as a tangent-space approximation. The objective $(r^\omega)^2/\sigma^2 + \log\sigma^2$ is the 1D Gaussian NLL of the scalar $r_i^\omega = \|\mathrm{Log}(\hat{R}_{1|t}^\top R_1)\|$. This is the tangent-space approximation of the true IGSO(3) NLL on the rotation residual; the approximation error is $O(\sigma^2)$ as $\sigma \to 0$ [Nikolayev and Savyolov, 1997]. For the concentration regime encountered in training ($\sigma_\omega \lesssim 0.5$ rad), the correction is small and a constant offset, which does not affect $\arg\min_\sigma \mathcal{L}_{\text{width}}$.

Proposition 1 (Self-regulating bridge width)**.** *For any fixed observed residual $r \geq 0$, the loss $\ell(r,\sigma)$ in Eq. (9) is a strictly convex function of $u = \sigma^2$ on $(0,\infty)$ and has a unique minimizer*

$$\sigma^*(r) \;=\; r, \qquad \ell(r,\sigma^*) \;=\; 1 + \log r^2 + \textit{const}.$$

Hence in the unconstrained limit, the pointwise minimizer of $\mathcal{L}_{\text{width}}$ (10) satisfies $\sigma_{\cdot,i}^*(t) = r_i^{\cdot}(t)$ almost surely in the data distribution: the optimal bridge width at (t,i) equals the drift model's residual magnitude at (t,i).

Proof. Let $u = \sigma^2 > 0$, so $\ell = r^2/u + \log u$. Differentiating, $d\ell/du = -r^2/u^2 + 1/u = (u - r^2)/u^2$. This vanishes uniquely at $u^* = r^2$. The second derivative is $d^2\ell/du^2 = 2r^2/u^3 - 1/u^2$. At $u^* = r^2$, $d^2\ell/du^2 = 2/r^4 - 1/r^4 = 1/r^4 > 0$, so u^* is the strict global minimizer. Taking square roots, $\sigma^* = r$. The summation over (i,t) in Eq. (10) decouples pointwise, so the global minimizer is pointwise the local minimizer. □

Corollary 1 (Bayesian posterior variance interpretation)**.** *Assume the drift is Bayes-optimal, i.e. $\hat{x}_{1|t} = \mathbb{E}[x_1 \mid x_t, a]$ and likewise for rotations. Then the minimizer of Eq. (10) in expectation satisfies*

$$\mathbb{E}\big[(\sigma_{x,i}^*)^2\big] \;=\; \mathbb{E}\big[(r_i^x)^2\big] \;=\; \mathrm{tr}\big(\mathrm{Cov}(x_{i,1} \mid x_t, a)\big),$$

i.e., the learned per-residue bridge width squared tracks the trace of the posterior covariance of the clean position given the noisy observation and sequence. The rotation case is analogous in the tangent-space approximation.

Proof. By Prop. 1, $\sigma^* = r$ pointwise, so $\mathbb{E}[\sigma^{*2}] = \mathbb{E}[r^2] = \mathbb{E}[\|x_1 - \mathbb{E}[x_1 \mid x_t, a]\|^2] = \mathrm{trCov}(x_1 \mid x_t, a)$ by the definition of posterior covariance. □

This corollary is the theoretical heart of EnsFlow: *the learned heteroscedasticity is an estimator of the Bayesian posterior uncertainty of the clean structure given the current noisy state*. In particular, the learned bridge width is not an arbitrary temperature parameter; it inherits the clean-structure posterior from the drift model itself.

Why boundedness does not harm the optimum. The parameterization (7) constrains $\alpha \in [\alpha_{\min}, \alpha_{\max}]$ and thus $\sigma \in [\alpha_{\min}\sigma_0(t), \alpha_{\max}\sigma_0(t)]$. Provided the base scale $\sigma_0(t)$ is calibrated so that the unconstrained optimum $\sigma^* = r$ lies in the interior (i.e. typical $r/\sigma_0(t) \in (\alpha_{\min}, \alpha_{\max})$), the bounded problem has the same minimizer as the unconstrained one. We set $\lambda_x = 4.0$ Å and $\lambda_\omega = 0.5$ rad based on RNA backbone empirics (typical $r_x \approx 2$ Å, $r_\omega \approx 0.25$ rad at $t = 0.5$), so that $\alpha^* \approx 1$ sits near the center of $[0.1, 3.0]$.

4.4 BridgeWidthNet: architecture for the modulation field

BridgeWidthNet predicts the pair $(\alpha_i^x, \alpha_i^\omega) \in [\alpha_{\min}, \alpha_{\max}]^2$ for each residue. To guarantee inference-time well-definedness (no dependence on unobservable ground-truth), we condition only on quantities available at test time:

$$(\alpha_i^x, \alpha_i^\omega) \;=\; g_\phi\big(c_i,\ \Delta_i^x,\ \Delta_i^\omega,\ \bar{p}_i,\ t\big), \tag{11}$$

where $c_i \in \mathbb{R}^{384}$ is the drift model's single-residue embedding; $\Delta_i^x = x_{i,t} - \hat{x}_{i,1|t}$ and $\Delta_i^\omega = \mathrm{Log}(\hat{R}_{i,1|t}^\top R_{i,t})$ are per-residue geometric residuals computed from the drift model's own clean-endpoint prediction (so the inputs are identical at train and test time); and $\bar{p}_i = \frac{1}{L}\sum_j p_{ij} \in \mathbb{R}^{128}$ is the column-mean of the pair representation, which encodes per-residue pairwise structural context.

The network is a lightweight 1D ConvNet with FiLM-style time conditioning [Perez et al., 2018]: the single-residue feature is linearly projected to a hidden dimension, added to a projection of $\bar{p}_i$, passed through four residual convolutional blocks (each internally modulated by a shared time embedding), and finally mapped through two independent linear heads and sigmoid bounding to $(\alpha_i^x, \alpha_i^\omega)$. The total parameter count is approximately 0.9M, a 6% overhead on top of the 14.2M parameter drift model. Further details are in App. B.

Why geometric residuals, not raw coordinates. Using $(\Delta_i^x, \Delta_i^\omega)$ rather than $(x_{i,t}, R_{i,t})$ has two benefits. First, Δ's are train/test consistent: they depend only on G_t and the drift model's prediction, never on the unobservable G_1. Second, the input semantics match what the output is doing: a large $\|\Delta\|$ means "the drift is still far from where it thinks the clean state is", which is exactly when the bridge should be wider.

4.5 Level 3: Experimental B-factor anchoring

Corollary 1 gives a *model-internal* interpretation of the learned bridge width: it estimates the model's own posterior uncertainty. Can we tie it to something experimentally measurable?

B-factors as physical flexibility. Crystallographic B-factors (isotropic atomic displacement parameters, $B = 8\pi^2\langle u^2\rangle$, where $\langle u^2\rangle$ is the mean-squared atomic displacement) are a physical measure of per-residue aleatoric flexibility. A stable base-paired stem has low B-factors; a mobile loop has high B-factors. If our learned α truly captures local flexibility, it should correlate with residue-level B-factors. Level 3 turns this implicit check into an explicit training signal.

Anchor loss. Let $\tilde{B}_i \geq 0$ be the residue-level B-factor for residue i after per-molecule normalization (log-transform, shift minimum to 0, scale mean to 1). We supervise the modulation factors with a simple MSE anchor:

$$\mathcal{L}_{\text{bfactor}} \;=\; \tfrac{1}{2L}\sum_{i=1}^{L}\Big[\big(\alpha_i^x - \tilde{B}_i\big)^2 + \big(\alpha_i^\omega - \tilde{B}_i\big)^2\Big], \tag{12}$$

and the Level 3 total loss is $\mathcal{L}_{\text{Level 3}} = \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{width}} + \lambda_B\,\mathcal{L}_{\text{bfactor}}$. We use $\lambda_B = 1$.

Why this is principled, not a patch. The NLL signal $\mathcal{L}_{\text{width}}$ is a statement about the model's *epistemic* uncertainty: "where do I make large drift errors". The B-factor signal $\mathcal{L}_{\text{bfactor}}$ is a statement about the system's *aleatoric* flexibility: "where does the molecule physically wiggle". For a well-trained model these should agree. Enforcing this agreement during training has two effects: (i) it gives the bridge width a well-posed target in early training, before the drift has converged enough for its residuals to be informative; (ii) it provides an independent falsifiability check—if α trained *without* B-factor supervision still correlates well with held-out B-factors, then the self-regulating signal is learning something real (this is our key qualitative experiment, §5.5).

Alternative formulation (discussion). The MSE anchor on α has a subtle unit asymmetry: $\tilde{B}$ is in units of $8\pi^2\langle u^2\rangle$ (after log and normalization), whereas α is a dimensionless multiplier on $\sigma_0(t)$. A more principled objective aligns $\log\sigma^2$ (or equivalently $\log\alpha$) with $\log\tilde{B}$, since $B \propto \langle u^2\rangle \propto \sigma^2$ under a harmonic assumption. We find empirically that the two are equivalent up to a reparameterization and report both in App. C.

4.6 Theoretical properties

Proposition 2 (Endpoint preservation)**.** *Under the parameterization Eqs. (5)–(7), the diffusion coefficient vanishes at both endpoints: $\sigma_i^x(0,\cdot) = \sigma_i^x(1,\cdot) = 0$ and $\sigma_i^\omega(0,\cdot) = \sigma_i^\omega(1,\cdot) = 0$ for all i, uniformly over any input to the network.*

Proof. $\sigma_0^x(0) = \lambda_x\sqrt{0\cdot 1} = 0$ and likewise $\sigma_0^x(1) = \lambda_x\sqrt{1\cdot 0} = 0$. The modulation α_i^x is bounded in $[\alpha_{\min}, \alpha_{\max}]$ by (7), so the product $\sigma_i^x = \sigma_0^x(t)\,\alpha_i^x$ is zero at $t \in \{0,1\}$ regardless of the network's output. The rotation case is identical. □

Proposition 3 (Degenerate limits)**.** *The Level 2 SDE has the following strict degenerate limits.*

(a) Level 1 ODE limit. *As $(\lambda_x, \lambda_\omega) \to (0,0)$ the SDE (3)–(4) converges to the deterministic ODE of Level 1 in the sense of strong convergence on any compact interval $[0, 1-\varepsilon]$: if $X^{(\lambda)}$ denotes the strong solution of Eq. (3) under drift u_θ and width $\lambda\sqrt{t(1-t)}\alpha$, and $X^{(0)}$ the ODE solution under the same drift, then for any Lipschitz drift and any $\varepsilon > 0$,*

$$\sup_{t\in[0,1-\varepsilon]} \mathbb{E}\big\|X_t^{(\lambda)} - X_t^{(0)}\big\|^2 \;\leq\; C_\varepsilon\,\lambda^2 \;\xrightarrow{\lambda\to 0}\; 0.$$

(b) Fixed-width FoldFlow-SFM limit. *If $\alpha_i^x(G_t, a, t) \equiv 1$ and $\alpha_i^\omega(G_t, a, t) \equiv 1$, the bridge width collapses to the global time-dependent schedule $(\sigma_0^x(t), \sigma_0^\omega(t))$ shared across all residues and independent of state. The resulting SDE is the fixed-width* $\mathrm{SE}(3)$ *Brownian bridge underlying FoldFlow-SFM [Bose et al., 2024].*

(c) Level 3 → Level 2. *Setting $\lambda_B = 0$ recovers Level 2 exactly.*

Proof sketch. (a) Standard vanishing-noise SDE-to-ODE convergence theorem. Writing the bridge SDE as $dX_t = u(X_t, t)\,dt + \lambda\, g(X_t, t)\,dW_t$ with $g = \sqrt{t(1-t)}\,\alpha$ bounded, the Itô isometry plus Grönwall's inequality yields $\mathbb{E}\|X_t^{(\lambda)} - X_t^{(0)}\|^2 \leq C(\varepsilon, u, g)\lambda^2$ for $t \in [0, 1-\varepsilon]$; see, e.g., Kloeden and Platen [1992, Ch. 5]. (b) Is immediate from the definition (5). (c) Is trivial. □

Remark 1 (Sampling well-definedness). The velocity (1) has a $1/(1-t)$ factor that is singular at $t = 1$. In practice we integrate the SDE on $[0, 1-\varepsilon]$ for small ε and take a final deterministic step from $(1-\varepsilon)$ to 1 using the drift model's clean-endpoint prediction. This mirrors the standard treatment in FoldFlow [Bose et al., 2024] and RNA-FrameFlow [Anand et al., 2024].

4.7 Inference: Euler–Maruyama on $\mathrm{SE}(3)^L$

At inference we sample $G_0 \sim \text{prior}$ (isotropic Gaussian for translation, uniform for rotation) and integrate the SDE forward in time via Euler–Maruyama:

$$x_{i,t+\Delta t} = x_{i,t} + \Delta t \cdot u_\theta^x(i,t) + \sqrt{\Delta t}\, \sigma_i^x(t, G_t, a)\, \xi_i^x, \quad \xi_i^x \sim \mathcal{N}(0, I_3), \tag{13}$$

$$R_{i,t+\Delta t} = R_{i,t} \cdot \mathrm{Exp}\Big(\Delta t \cdot u_\theta^\omega(i,t) + \sqrt{\Delta t}\, \sigma_i^\omega(t, G_t, a)\, \xi_i^\omega\Big), \quad \xi_i^\omega \sim \mathcal{N}(0, I_3). \tag{14}$$

Each step requires one drift forward (producing $\hat{G}_{1|t}$ and the velocities) and one BridgeWidthNet forward (producing $(\alpha^x, \alpha^\omega)$ from the drift's residuals). The final step takes a deterministic step to $t = 1$.

5 Experiments

Status note. *This section is the main work-in-progress block of the draft. We state the intended experimental protocol in full detail; Tables 2 and 3 show the empty result shells ready to be populated after the training runs complete. All metric definitions match our implementation in* `rna_backbone_design/analysis/metrics.py`.

5.1 Datasets

RNA conformational ensemble dataset. We build an RNA ensemble dataset of **[TODO: N]** sequence clusters, where each cluster contains multiple experimentally determined or MD-sampled conformations of the same or similar sequences. Clusters are formed by sequence similarity with identity threshold **[TODO: τ]**. For each cluster, the ground-truth ensemble consists of the K distinct conformers within the cluster. Data sources include: (i) multi-state PDB entries (NMR ensembles and multi-chain asymmetric units), (ii) **[TODO: RNA MD simulation data source]**, and (iii) **[TODO: homologous structures of the same RNA family]**. We split at the cluster level so no conformer from a training cluster appears in validation or test. **[TODO: Report sizes and splits.]**

B-factor data (Level 3 only). For each training structure we extract residue-level B-factors by averaging the heavy-atom isotropic B values per residue, taking the log, and per-molecule normalizing to zero-minimum, unit-mean (matching `fk_utils.normalize_b_factors_positive`). Structures without experimental B-factors (e.g. NMR) are held out of Level 3 training.

5.2 Metrics

All metrics are computed on C4′ atom positions after per-sample Kabsch [Kabsch, 1976] superimposition, following our reference implementation.

Single-structure quality. For each test sequence we generate $G =$ **[TODO: 50]** samples and report *best-of-G* RMSD and TM-score against each ground-truth conformer (`compute_single_metrics`).

Ensemble coverage and precision. Let $\{g_j\}_{j=1}^{G}$ be the generated ensemble and $\{r_k\}_{k=1}^{K}$ be the ground-truth ensemble for a given test cluster. We report, as in `compute_ensemble_metrics`:

$$\text{AMR-R} = \frac{1}{K}\sum_k \min_j \text{RMSD}(g_j, r_k) \qquad \text{(recall)},$$

$$\text{AMR-P} = \frac{1}{G}\sum_j \min_k \text{RMSD}(g_j, r_k) \qquad \text{(precision)},$$

$$\text{COV-R}_\delta = \frac{1}{K}\sum_k \mathbf{1}\left[\min_j \text{RMSD}(g_j, r_k) \le \delta\right] \qquad \text{(recall@}\delta\text{)},$$

$$\text{COV-P}_\delta = \frac{1}{G}\sum_j \mathbf{1}\left[\min_k \text{RMSD}(g_j, r_k) \le \delta\right] \qquad \text{(precision@}\delta\text{)},$$

for $\delta \in \{3, 4\}$ Å. We additionally report mean *pairwise RMSD* within $\{g_j\}$ as an unsupervised diversity measure.

Physical validity. Per generated sample we compute (`calc_rna_c4_c4_metrics`): C4′-C4′ bond-length deviation from canonical, valid-bond percentage, number of inter-residue C4′ clashes (threshold 1.0 Å), and radius of gyration. We report aggregated means.

Uncertainty–flexibility correlation. This is the key qualitative metric. For each test structure we extract the learned per-residue α_i^x and α_i^ω along the sampling trajectory (averaged over timesteps, and over the generated ensemble) and compute Pearson and Spearman correlations against held-out residue-level B-factors. *This is computed for Level 2 (no B-factor supervision) and Level 3 separately*; a high correlation in Level 2 is the cleanest evidence that the self-regulating NLL learns a physically meaningful uncertainty field.

5.3 Baselines

We compare against four baseline classes that together test each of our contributions:

- **Deterministic flow matching** (RNA-FrameFlow-style Level 1, our re-implementation). Tests whether any stochasticity at all is needed.
- **Fixed-σ bridge** (Level 2 with $\alpha \equiv 1$). Tests whether *learned* heteroscedasticity beats fixed global-in-time stochasticity—this is the closest analogue of FoldFlow-SFM.
- **RNA-FrameFlow with independent repeated sampling.** Off-the-shelf single-structure generator; repeated sampling to form an "ensemble". Tests whether training for ensemble generation at all helps, versus decoding-time diversity.
- **AlphaFold3 ensemble** (where applicable). Off-the-shelf structure predictor run multiple times with different random seeds as a strong real-world baseline.

5.4 Main results

[TODO: Write-up of what the main results show. Expected findings: (i) Level 1 underperforms on coverage due to mode collapse; (ii) fixed-σ gains coverage but loses precision; (iii) Level 2 beats both on the coverage–precision Pareto curve; (iv) Level 3 further improves the Spearman correlation ρ_B without sacrificing coverage.]

Table 2: Main results on the RNA conformational ensemble benchmark. Lower AMR and higher COV indicate better ensemble match; higher pairwise RMSD indicates more diversity. ρ_B is Spearman correlation of learned $(\alpha^x + \alpha^\omega)/2$ with held-out B-factors. Values are **[TODO: placeholders to be filled after training runs]**.

Method	Coverage@δ (R/P, %)		AMR (Å, R/P)		Pairwise RMSD (Å)	Clash	ρ_B Spearman
	$\delta = 3$	$\delta = 4$	Recall	Precision			
Level 1 (deterministic)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	–
Fixed-σ bridge	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	–
RNA-FrameFlow (repeated)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	–
AlphaFold3 ensemble	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	–
EnsFlow Level 2 (ours)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**
EnsFlow Level 3 (ours)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**

Table 3: Ablation studies. Each row isolates a single design decision. **[TODO: Values to be filled.]**

Variant	AMR-R (Å)	COV-R@4	Pairwise RMSD	ρ_B	Comment
Full EnsFlow L2	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	reference
$\alpha \equiv 1$ (no learned modulation)	**[TODO:]**	**[TODO:]**	**[TODO:]**	–	Prop. 3(b)
$\lambda_x, \lambda_\omega \to 0$ (no noise)	**[TODO:]**	**[TODO:]**	**[TODO:]**	–	Prop. 3(a)
No pair input $\bar{p}_i$	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	removes cluster signal
No geometric residual Δ	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	raw G_t input instead
Unbounded α (softplus)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	stability check
Half λ base scale	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	sensitivity to σ_0
Double λ base scale	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	sensitivity to σ_0
Full EnsFlow L3 (+ B-factor)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	reference
$\lambda_B \to 0$ (= L2)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**	isolates B-anchor

5.5 Ablations

The headline ablation. The most important row is *"Full EnsFlow L2"*: L2 has no B-factor supervision whatsoever, and yet we measure ρ_B—the Spearman correlation of its learned α with held-out B-factors. A significantly positive ρ_B for L2 is the cleanest possible evidence that the self-regulating NLL is not just fitting residual noise but learning a physically meaningful uncertainty field (Cor. 1). **[TODO: Report the actual number and a scatter plot.]**

Sensitivity to σ_0. By Prop. 1, the optimum is $\sigma^* = \|r\|$ in the unconstrained limit; the bounded parameterization (7) clips this to $[\alpha_{\min}\sigma_0(t), \alpha_{\max}\sigma_0(t)]$. If σ_0 is mis-calibrated (too small $\Rightarrow$ α saturates at $\alpha_{\max}$; too large $\Rightarrow$ saturates at $\alpha_{\min}$), the model loses modeling capacity. The half- and double-λ ablations test robustness to this choice. We expect performance to be essentially flat for $\lambda \in [\lambda^*/2, 2\lambda^*]$ and degrade outside.

5.6 Qualitative: uncertainty field visualization

[TODO: Figure: heatmap of learned α_i^x along the sequence for a test structure, alongside (a) experimental B-factors, (b) the RNA secondary structure (stems vs loops). Expectation: α tracks B-factors, lights up in loops and junctions, stays low in stems.]

6 Discussion and Conclusion

We have introduced EnsFlow, a three-level hierarchical framework for RNA conformational ensemble generation on $\mathrm{SE}(3)^L$. The methodological core is a conditional heteroscedastic stochastic bridge whose per-residue width field is trained by a heteroscedastic NLL of the drift model's own residuals, with a closed-form optimum $\sigma^* = \|r\|$ that admits a Bayesian posterior-variance interpretation (Cor. 1). We further show that experimental B-factors can be used as a direct physical anchor for the learned uncertainty, giving a concrete falsifiability check for the self-regulating signal. The three levels are connected by strict degenerate limits (Prop. 3), making the framework a clean generalization of both deterministic $\mathrm{SE}(3)$ flow matching and fixed-width stochastic bridges.

Limitations. Our rotation NLL uses a tangent-space Gaussian approximation to the IGSO(3) density, exact only in the small-σ limit. The Bayesian interpretation (Cor. 1) requires a Bayes-optimal drift, which is an idealization. Our B-factor anchor assumes harmonic flexibility; anharmonic or multi-state flexibility may be better captured by richer priors (e.g. anisotropic displacement parameters). We target RNA; transfer to proteins is straightforward but not evaluated.

Broader impact. RNA conformational ensemble generation is a fundamental tool for RNA drug discovery and mechanistic biology. Better ensemble models may accelerate the design of RNA-targeted therapeutics (riboswitch modulators, ribozyme engineering) and our understanding of regulatory RNAs. We see no immediate dual-use risks beyond those general to structural biology.

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A Proofs

Prop. 1 (detailed). Already given in the main text. We only note that the result is an instance of the MLE for the scale parameter of a zero-mean 1D Gaussian from a single observation $|r|$; the convexity in $u = \sigma^2$ follows because $r^2/u + \log u$ has second derivative $2r^2/u^3 - 1/u^2$ which is positive at $u^* = r^2$.

Cor. 1 (detailed). The Bayes-optimal denoiser is $\hat{x}_{1|t} = \mathbb{E}[x_1 \mid x_t, a]$. The residual is $r^x = x_1 - \hat{x}_{1|t}$, and

$$\begin{aligned}\mathbb{E}[\|r^x\|^2] &= \mathbb{E}[\|x_1 - \mathbb{E}[x_1 \mid x_t, a]\|^2] = \mathbb{E}\big[\mathbb{E}\big[\|x_1 - \mathbb{E}[x_1 \mid x_t, a]\|^2 \mid x_t, a\big]\,\big] \\ &= \mathbb{E}\big[\mathrm{trCov}(x_1 \mid x_t, a)\big],\end{aligned}$$

where the last equality is the definition of the trace of the conditional covariance. Combining with Prop. 1 yields $\mathbb{E}[(\sigma^*)^2] = \mathbb{E}\mathrm{trCov}(x_1 \mid x_t, a)$. For the rotation case, replace $x_1 - \mathbb{E}[x_1 \mid x_t, a]$ by $\mathrm{Log}(\mathbb{E}[R_1 \mid x_t, a]^\top R_1)$, interpreted in the tangent-space Gaussian limit, where the posterior mean is taken in the intrinsic Fréchet sense; the remainder of the calculation is formally identical.

Prop. 2 (detailed). Already given in the main text.

Prop. 3(a) (detailed). We invoke the vanishing-noise SDE-to-ODE convergence theorem in the form of Kloeden and Platen [1992, Ch. 10, Thm. 2.2]. Write $dX_t^{(\lambda)} = u(X_t^{(\lambda)}, t)\,dt + \lambda\, g(X_t^{(\lambda)}, t)\,dW_t$ and $dX_t^{(0)} = u(X_t^{(0)}, t)\,dt$. Under the standard assumptions that u is globally Lipschitz in x and continuous in t on $[0, 1-\varepsilon]$, and that g is bounded on the same set (which holds because $g = \sqrt{t(1-t)}\alpha$ with $\alpha \le \alpha_{\max}$ bounded), the Itô isometry gives

$$\mathbb{E}\|X_t^{(\lambda)} - X_t^{(0)}\|^2 \le C_u \int_0^t \mathbb{E}\|X_s^{(\lambda)} - X_s^{(0)}\|^2\,ds + \lambda^2 \int_0^t \mathbb{E}\|g(X_s^{(\lambda)}, s)\|^2\,ds. \tag{15}$$

By Grönwall's inequality, the first term yields an exponential factor in t and the second term is bounded by $C_g\,\lambda^2\,T$. Taking $T = 1 - \varepsilon$ and combining constants into C_ε yields $\sup_{t\in[0,1-\varepsilon]} \mathbb{E}\|X_t^{(\lambda)} - X_t^{(0)}\|^2 \le C_\varepsilon \lambda^2 \to 0$ as $\lambda \to 0$. This establishes strong convergence. The rotation case is handled analogously by lifting to a local chart on $\mathrm{SO}(3)$ and using the fact that the exponential map is a local diffeomorphism.

Prop. 3(b),(c) (detailed). Immediate from the definitions.

B Architectural details for BridgeWidthNet

BridgeWidthNet is a 1D ConvNet with FiLM [Perez et al., 2018] time conditioning. Let $H = 128$ be the hidden dimension.

Time conditioning (shared). The timestep $t \in [0, 1]$ is sinusoidally embedded to $\mathbb{R}^{64}$ and projected through a two-layer MLP to produce a shared conditioning vector $\mathbf{c}_{\text{time}} \in \mathbb{R}^H$.

Single-residue path. We concatenate the drift model's residue embedding $c_i \in \mathbb{R}^{384}$ with the two geometric residuals $\Delta_i^x \in \mathbb{R}^3$ and $\Delta_i^\omega \in \mathbb{R}^3$ to form a 390-dimensional feature, projected by a two-layer MLP to $\mathbb{R}^H$.

Pair path. The column-mean of the pair representation, $\bar{p}_i = \frac{1}{L}\sum_j p_{ij} \in \mathbb{R}^{128}$, is linearly projected to $\mathbb{R}^H$ and added to the single-residue feature. This fuses single-residue context with the drift model's pairwise structural view.

Backbone. Four residual 1D convolutional blocks (kernel size 5, dilation 1). Each block is: LayerNorm $\to$ Conv1D $\to$ FiLM($\mathbf{c}_{\text{time}}$) $\to$ SiLU $\to$ Conv1D $\to$ residual sum. The FiLM layer applies an affine modulation $h \mapsto \gamma(\mathbf{c}_{\text{time}}) \odot h + \beta(\mathbf{c}_{\text{time}})$.

Output heads. Two independent linear heads produce the raw logits $(u_i^x, u_i^\omega) \in \mathbb{R}^2$, passed through $\alpha = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\sigma_{\text{sig}}(u)$ with $(\alpha_{\min}, \alpha_{\max}) = (0.1, 3.0)$.

Parameter count. $\approx$ 0.9M: 24.8K (time MLP), 66.6K (single projection), 16.5K (pair projection), 789.5K (four ResConv blocks), 0.8K (heads). This is a 6.3% overhead on top of the 14.2M-parameter drift model.

C B-factor normalization and anchor loss details

Normalization. Given residue-level B-factors $B_i \geq 0$ obtained by averaging heavy-atom isotropic B-values per residue, we apply

$$\tilde{B}_i = \frac{\log(1+B_i) - \min_j \log(1+B_j)}{\text{mean}_k(\log(1+B_k) - \min_j \log(1+B_j))},$$

giving $\tilde{B}_i \geq 0$ with mean 1 per molecule.

Alternative: log-sigma alignment. A more principled target is $\mathcal{L}_{\text{bfactor}}^{\log} = \frac{1}{L}\sum_i (\log \sigma_{x,i}^2 - \log \tilde{B}_i^2)^2$, which respects the harmonic relation $B \propto \sigma^2$. We find empirically that this and the direct MSE on α used in Eq. (12) are equivalent up to a reparameterization; we report the simpler form in the main text and include a comparison in **[TODO: Appendix table]**.

D Hyperparameters and training details

Table 4: Hyperparameters (drift model follows RNA-FrameFlow defaults).

Parameter	Value
Drift model	IPA, 5 blocks, $c_s = 256$, $c_z = 128$, 14.2M params
Flow weights (w_x, w_ω)	$(2.0,\ 1.0)$
BridgeWidthNet hidden dim H	128
BridgeWidthNet blocks	4
Base schedule λ_x, λ_ω	4.0 Å, 0.5 rad
Modulation bounds $(\alpha_{\min}, \alpha_{\max})$	$(0.1,\ 3.0)$
B-factor weight λ_B	1.0
Drift optimizer LR	1×10^{-5} (Level 2/3, post-L1 pretraining)
BridgeWidthNet optimizer LR	3×10^{-4}
Batch size	10
Max sequence length	300
Inference SDE steps	60

E Experiment plan and milestone checklist

Milestone 1: Level 1 baseline reproduction. Train the deterministic Level 1 model on the RNA ensemble dataset to convergence; verify single-structure metrics match the RNA-FrameFlow reported numbers within tolerance.

Milestone 2: Level 2 training (self-regulating NLL). Starting from the Level 1 checkpoint, freeze the drift to a low learning rate ($\lambda_{\text{drift}} = 1 \times 10^{-5}$), add BridgeWidthNet with lr 3×10^{-4}, and train to convergence of $\mathcal{L}_{\text{width}}$. Monitor `train/bridge_r_x_mean`, `train/bridge_r_w_mean`, `train/bridge_alpha_x_mean`, `train/bridge_alpha_w_mean` to recalibrate $\lambda_x, \lambda_\omega$ after the first epoch so that the typical optimal $\alpha^* \approx 1$.

Milestone 3: Level 3 training (B-factor anchor). Add $\mathcal{L}_{\text{bfactor}}$ with $\lambda_B = 1$ and continue training. Monitor ρ_B (Spearman of α against held-out B-factors).

Milestone 4: Baselines. Train the fixed-σ bridge ablation ($\alpha \equiv 1$) and the RNA-FrameFlow repeated-sampling baseline; run AlphaFold3 on the test set with multiple seeds.

Milestone 5: Evaluation. For each model, generate $G = 50$ samples per test sequence; compute all metrics from §5.2. Produce Tables 2 and 3.

Milestone 6: Qualitative figures. Scatter plot of α vs. B-factor (Level 2 and Level 3); heatmap of α along the sequence on a representative test structure with secondary structure annotation.

Compute budget (estimate). Drift (Level 1): **[TODO: x]** A100-days. Level 2 fine-tuning: **[TODO: y]** A100-days. Level 3 fine-tuning: **[TODO: z]** A100-days. Baselines and evaluation: **[TODO: w]** A100-days. Total: **[TODO: $\approx$ A100-days]**.